# AQUAMESH

EVERY NODE MONITORED

Presented by

**HydroHive**

# PROBLEMS

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### *URBAN WATER DISTRIBUTION SYSTEMS FACE:*

- Undetected leaks
- Pressure instability
- pH imbalance affecting water quality
- Reactive maintenance after failure

### *THIS RESULTS IN:*

- Water wastage
- Infrastructure damage
- Increased energy usage
- Delayed intervention

OBJECTIVES

1 Models a pipeline network using graph theory

2 Simulates pressure, pH, and leak

3 Trains an AI model on normal operating conditions

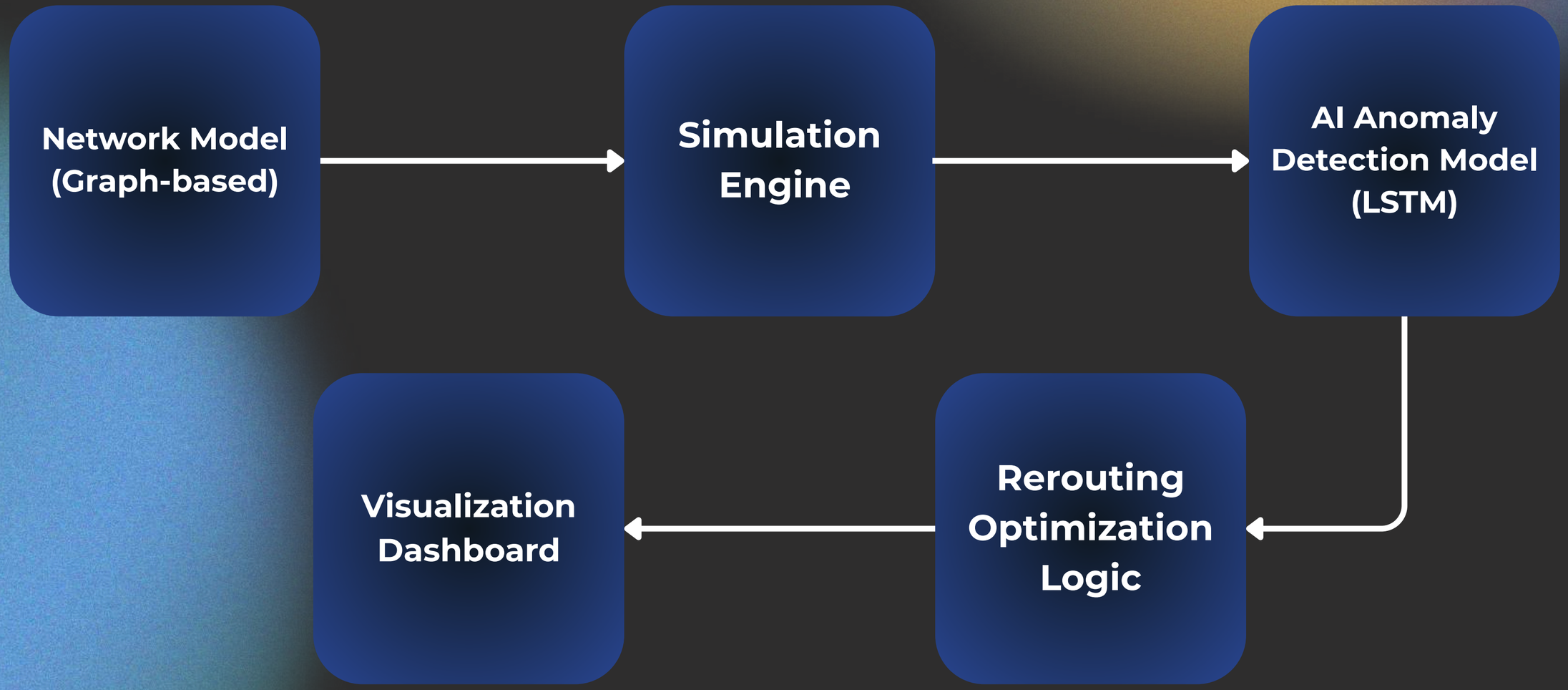
4 Detects anomalies automatically

5 Highlights affected junctions

6 Suggests rerouting strategies to minimize water loss

# OBJECTIVES

# WORKFLOW

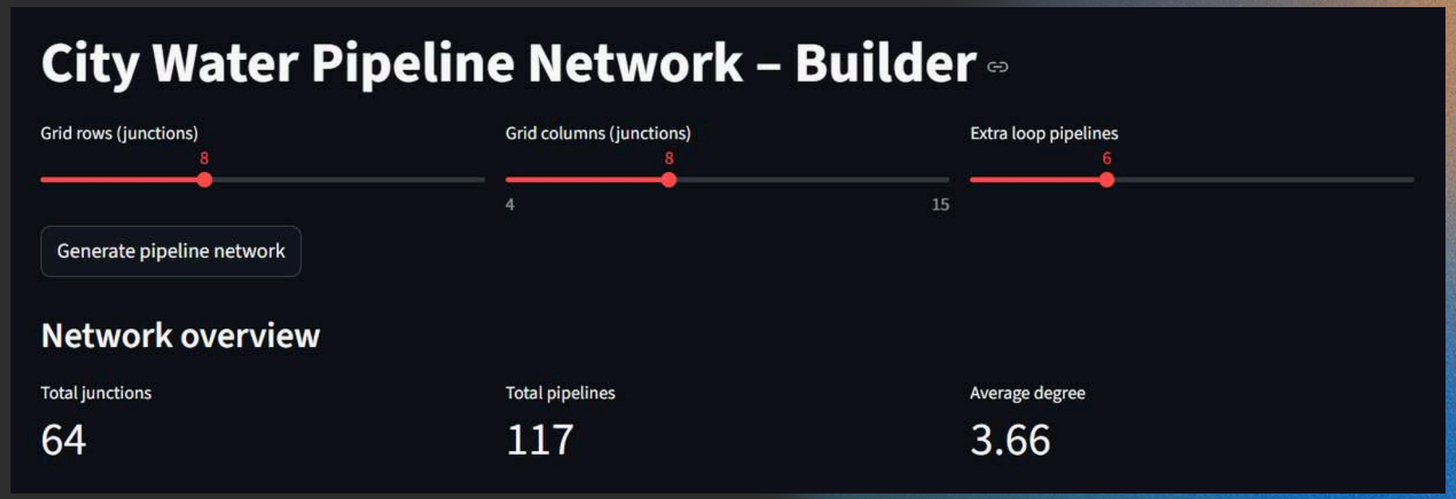

Network Model (Graph-based)
Simulation Engine
AI Anomaly Detection Model (LSTM)
Rerouting Optimization Logic
Visualization Dashboard

STEP 1

# PIPELINE NETWORK CREATION

***We construct a grid-based graph:***

- *Nodes → Junction points*
- *Edges → Pipelines*
- *Each node stores:*
  - *Pressure*
  - *pH*
  - *Leak status*
  - *Abnormal state*

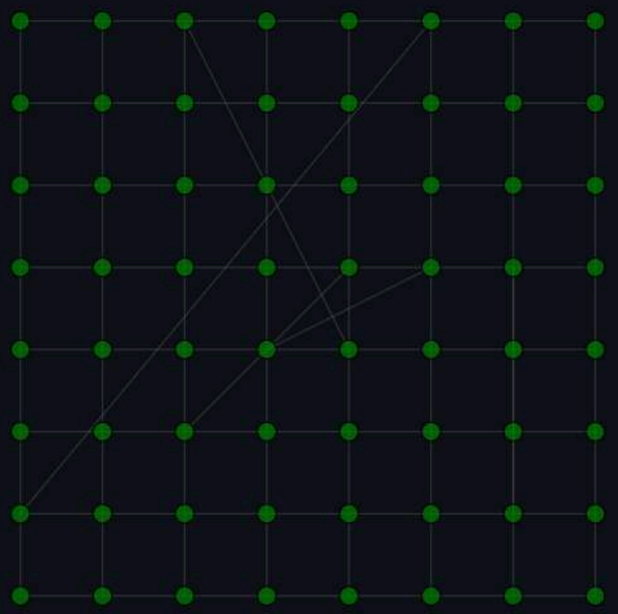



***This forms the digital twin of a water distribution network.***

Deploy

# Structural view (pipeline layout sketch)

**City Water Pipeline Network – Tick 0**



# SIMULATION STEP 2

## Hydraulic & Quality Simulation – Pipeline Network

Simulation ticks 60

Leak probability per junction 0.04

pH anomaly probability 0.04

Pressure anomaly probability 0.03

Run simulation

### Simulation snapshot

Each row represents a junction at a specific tick with pressure, pH, neighbor statistics, and labels.

| | junction_id | tick | x | y | pressure | pH | degree | neighbor_avg_pressure | neighbor_avg_pH | delta_pressure_from_baseline | delta_pH_from_base |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | J-0-0 | 0 | 0 | 7 | 49.275 | 7.1951 | 2 | 50.6808 | 7.1549 | -0.725 | -0.0 |
| 1 | J-0-1 | 0 | 1 | 7 | 49.8945 | 7.226 | 3 | 49.8115 | 7.1655 | -0.1055 | 0. |
| 2 | J-0-2 | 0 | 2 | 7 | 47.8137 | 7.1334 | 4 | 51.4088 | 7.2131 | -2.1863 | -0.0 |
| 3 | J-0-3 | 0 | 3 | 7 | 52.0466 | 7.2237 | 3 | 45.5207 | 7.1927 | 2.0466 | 0.0 |
| 4 | J-0-4 | 0 | 4 | 7 | 37.7005 | 7.2625 | 3 | 50.2583 | 7.2155 | -12.2995 | 0.0 |
| 5 | J-0-5 | 0 | 5 | 7 | 48.1065 | 7.1741 | 4 | 43.2624 | 7.1943 | -1.8935 | -0.0 |
| 6 | J-0-6 | 0 | 6 | 7 | 36.1243 | 7.1611 | 3 | 49.5095 | 7.2163 | -13.8757 | -0.0 |
| 7 | J-0-7 | 0 | 7 | 7 | 49.6933 | 7.2324 | 2 | 44.1041 | 7.224 | -0.3067 | 0.0 |
| 8 | J-1-0 | 0 | 0 | 6 | 51.4671 | 7.0838 | 3 | 46.9457 | 7.1889 | 1.4671 | -0.1 |
| 9 | J-1-1 | 0 | 1 | 6 | 52.3457 | 7.1681 | 4 | 51.0197 | 7.1552 | 2.3457 | -0.0 |

***The simulation runs over time steps:***

*Normal pressure fluctuations*

*Normal pH variation*

*Random leak generation*

*Random abnormal pressure spikes*

***Pressure and pH are influenced by neighboring nodes to mimic real-world pipeline propagation behavior. This produces realistic time-series data.***

*The LSTM model is trained on:*

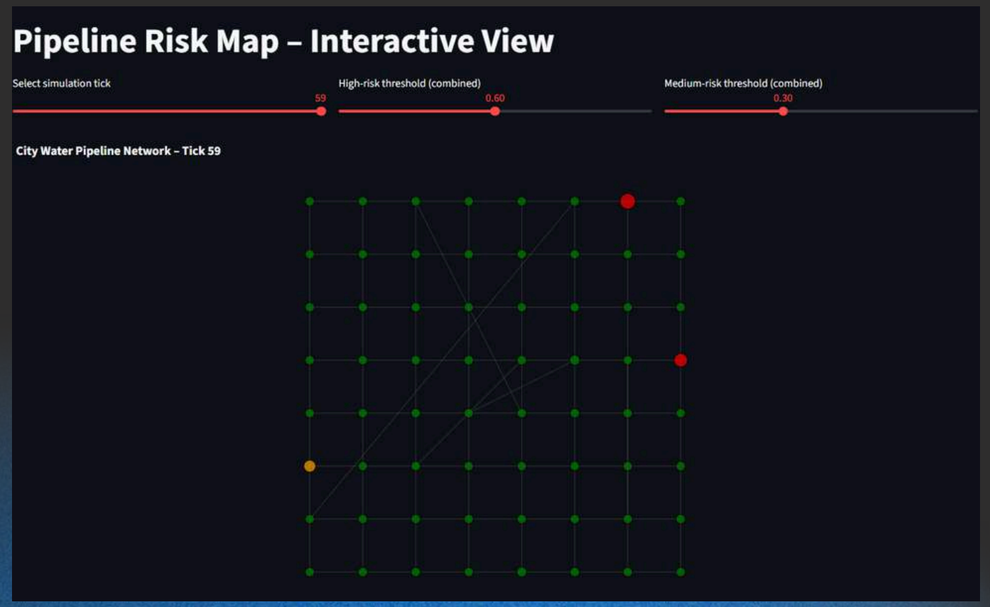
| | |
|---|---|
| Sequences of pressure values | Sequences of pH values |

*When deviation occurs:*

| | |
|---|---|
| The model flags the junction as anamalous | Assigns a risk score |

# AI TRAINING STEP 3

# STEP 4




# ANOMALY DETECTION & RISK SCORING

**1**

*The system identifies affected junctions*

**2**

*Highlights them in the network*

**3**

*Computes an anomaly score*

**4**

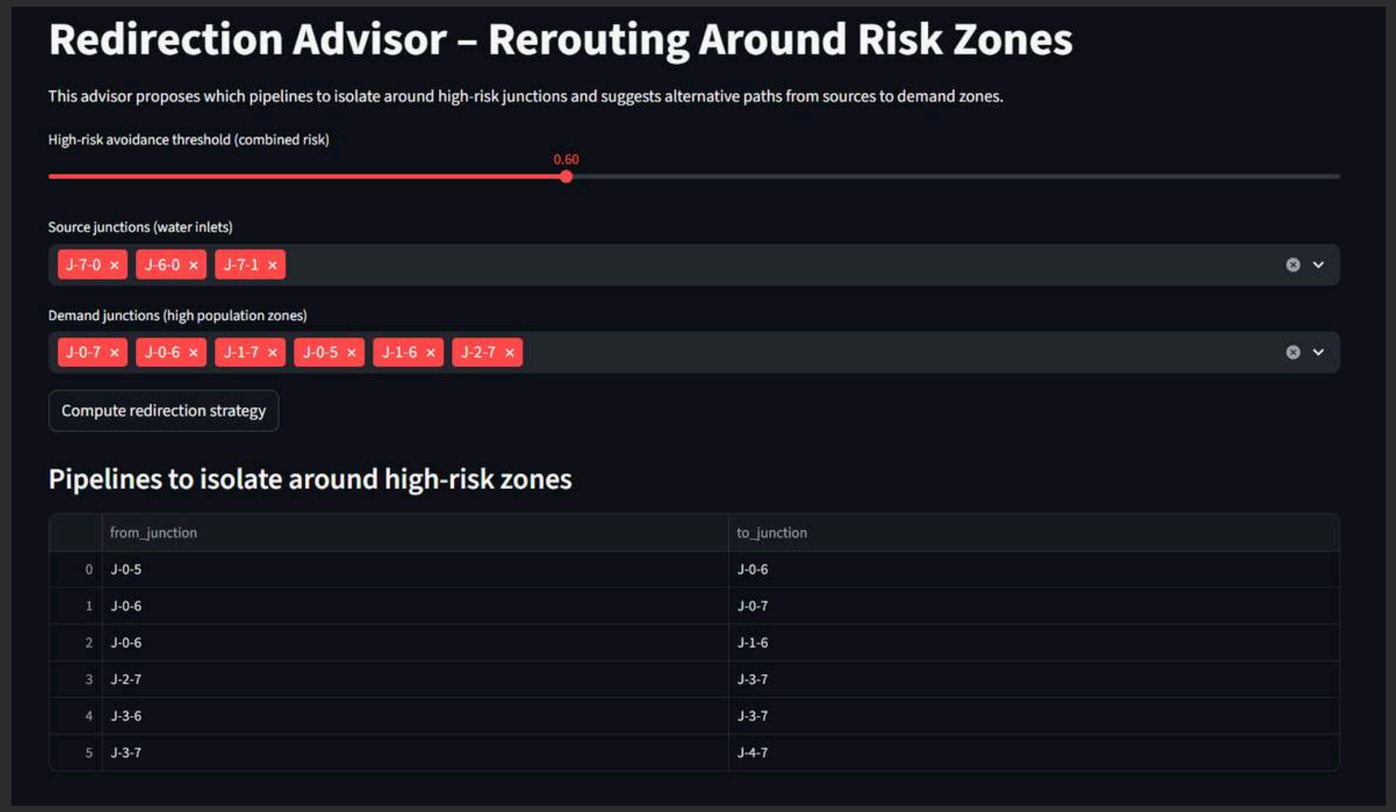
*Labels severity level*

# Top risk junctions (latest predictions)

Number of junctions to list

15

| | junction_id | pressure | pH | leak_risk | abnormal_risk | combined_risk | ground_truth |
|---|---|---|---|---|---|---|---|
| 6 | J-0-6 | 51.7426 | 7.6526 | 0.0001 | 0.9705 | 0.9705 | Abnormal |
| 31 | J-3-7 | 50.992 | 7.4025 | 0.00008 | 0.6708 | 0.6708 | Abnormal |
| 40 | J-5-0 | 50.7763 | 7.3704 | 0.00007 | 0.4354 | 0.4354 | Abnormal |
| 29 | J-3-5 | 49.5155 | 7.3507 | 0.0001 | 0.1439 | 0.1439 | Normal |
| 60 | J-7-4 | 50.0267 | 7.3417 | 0.0001 | 0.1078 | 0.1078 | Normal |
| 27 | J-3-3 | 48.5292 | 7.1847 | 0.0003 | 0.0398 | 0.0398 | Normal |
| 57 | J-7-1 | 50.1177 | 7.2949 | 0.0003 | 0.035 | 0.035 | Normal |
| 28 | J-3-4 | 49.9159 | 7.1374 | 0.0001 | 0.0292 | 0.0292 | Normal |
| 37 | J-4-5 | 48.8563 | 7.0901 | 0.0002 | 0.0276 | 0.0276 | Normal |
| 38 | J-4-6 | 51.212 | 7.3148 | 0.00004 | 0.0229 | 0.0229 | Normal |




# REROUTING STEP 5

***After detecting a leak:***

- The system searches alternate paths in the graph
- Suggests redirecting flow
- Reduces load on the affected pipeline

***This simulates how water authorities could respond immediately to minimize loss.***

## Pipelines to isolate around high-risk zones

| | from_junction | to_junction |
|---|---|---|
| 0 | J-0-5 | J-0-6 |
| 1 | J-0-6 | J-0-7 |
| 2 | J-0-6 | J-1-6 |
| 3 | J-2-7 | J-3-7 |
| 4 | J-3-6 | J-3-7 |
| 5 | J-3-7 | J-4-7 |

## Recommended rerouted paths

| | demand_junction | source_junction | path_sequence | weighted_length | explanation |
|---|---|---|---|---|---|
| 0 | J-0-7 | J-6-0 | J-6-0 → J-0-5 → J-1-5 → J-1-6 → J-1-7 → J-0-7 | 5.28 | Reroute demand junction J-0-7 from source J-6-0 using a path that bypasses high-risk z |
| 1 | J-0-6 | J-6-0 | J-6-0 → J-0-5 → J-0-6 | 101.06 | Reroute demand junction J-0-6 from source J-6-0 using a path that bypasses high-risk z |
| 2 | J-1-7 | J-6-0 | J-6-0 → J-0-5 → J-1-5 → J-1-6 → J-1-7 | 4.19 | Reroute demand junction J-1-7 from source J-6-0 using a path that bypasses high-risk z |
| 3 | J-0-5 | J-6-0 | J-6-0 → J-0-5 | 1.06 | Reroute demand junction J-0-5 from source J-6-0 using a path that bypasses high-risk z |
| 4 | J-1-6 | J-6-0 | J-6-0 → J-0-5 → J-1-5 → J-1-6 | 3.13 | Reroute demand junction J-1-6 from source J-6-0 using a path that bypasses high-risk z |
| 5 | J-2-7 | J-6-0 | J-6-0 → J-0-5 → J-1-5 → J-2-5 → J-2-6 → J-2-7 | 5.14 | Reroute demand junction J-2-7 from source J-6-0 using a path that bypasses high-risk z |

# TECH STACK

1 **PYTHON**

2 **NETWORKX**
NETWORK MODELING

3 **NUMPY & PANDAS**
DATA SIMULATION

1 **LSTM**
TENSORFLOW / KERAS OR PYTORCH

2 **MATPLOTLIB/PLOTLY**
VISUALISATION

3 **STREAMLIT**
DASHBOARD DEPLOYMENT

TECH STACK

# FUTURE SCOPE

***Future improvements:***

- Train on larger datasets
- Predict anomalies before they happen
- Integrate real IoT sensor data
- Add energy optimization modeling

***Long-term vision:***

*Predictive and preventive water infrastructure management*

# THANK YOU

***Presented by***
Sangamithira J
Shruthi R
Jessica J
Nandhakishore M